

FINAL PROJECT – REINFORCEMENT LEARNING

Vision-Based Autonomous Navigation of a Differential-Drive Robot via Proximal Policy Optimization in MuJoCo

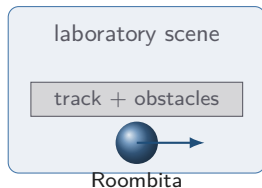
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Roombita RL Arena · MuJoCo · Stable-Baselines3

Why learn to navigate?

- Real robots act **continuously**, not on a grid
- They face obstacles, geometry, contact forces
- Classical planners need a **prior map**
- RL learns control **from interaction**
- Our goal: a **camera-to-wheels** policy



Follow the line, avoid clutter,
complete the loop.

From a discrete maze to a physical robot

Milestone 1



This work

- Tabular / VFA Q-Learning
- 5×5 GridWorld maze
- Discrete moves: N, S, E, W
- Value table, no perception

- PPO actor-critic + gSDE
- MuJoCo physics simulation
- Continuous **Twist** (v, ω)
- Vision via learned segmenter

A value table cannot represent continuous control from pixels.

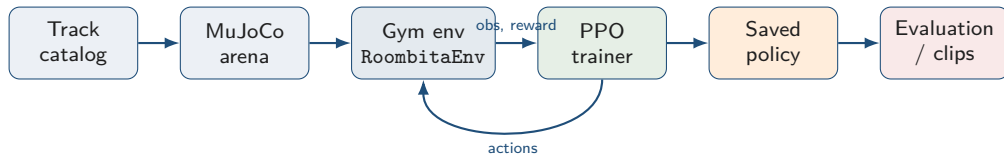
Contributions

- Continuous-control **MuJoCo** environment with a Twist interface
- **Two-stage perception** for sim-to-real transfer
- **Optimized PPO**: gSDE, frame-stacking, track curriculum
- Dense **arc-length reward** with anti-hacking safeguards
- **Ablation** separating the perception gap from the stability gap

PART 1

Method

System pipeline



- Modular: same agent runs across tracks and observation modes
- Evaluation can swap the learned segmenter for the ground-truth mask

Robot and control interface

- Differential drive + on-board camera
- Action is a **Twist**: $a_t = [v, \omega] \in [-1, 1]^2$
- Mapped to wheel velocities internally
- Exactly the command a **real** robot receives

v	ω	motion
+	0	forward
0	+	rotate in place
+	+	forward-left curve
+	-	forward-right curve

$$v_{\max}=0.6 \text{ m/s}, \quad \omega_{\max}=3.0 \text{ rad/s}$$

Perception: train on masks, deploy through a segmenter



- Policy **trained** on clean simulator masks (fast, stable)
- Frozen U-Net **bridges** RGB $\rightarrow$ mask at deployment
- Identical mask distribution at train and test time
- Control never sees raw pixels $\Rightarrow$ sim-to-real ready

Reward: signed progress plus safety

$$R_t = w_p \Delta s + r_{\text{safe}} + r_{\text{avoid}} + r_{\text{ret}} - r_{\text{col}} - r_{\text{off}} - r_{\text{stuck}} - r_{\text{smooth}}$$

- Dense **signed arc-length** progress drives learning
- Terminal events: arrival (+50), timeout, flip (−15)
- **Gated** return bonus removes an edge-oscillation exploit
- **Smoothness** penalty curbs jitter and roll-over

Optimized PPO configuration

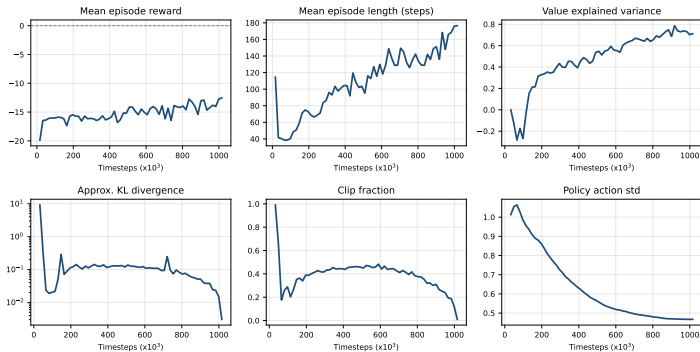
- **8 parallel envs**, six-track curriculum
- **gSDE** for smooth continuous exploration
- **4-frame stacking** adds motion cues
- **Linear LR decay** over 10^6 steps
- NatureCNN encoder, 256×256 heads

Param	Value
LR	2.5×10^{-4}
γ / λ	0.995 / 0.95
Clip ϵ	0.2
Rollout	16,384
Batch	256

PART 2

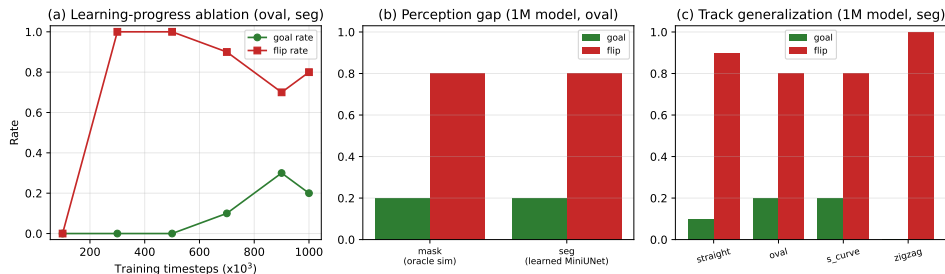
Results

Training is stable — reading the reward honestly



- Reward rises $-19.9 \rightarrow -12.6$: hard six-track *average*, not failure
- Explained variance $\rightarrow 0.71$; KL and clip-fraction settle; std shrinks
- Episodes grow longer — the agent survives and drives farther

Quantitative ablation



- (a) Competence emerges after 700k, **peaks at 900k**
- (b) Perception gap ≈ 0 : learned seg $\approx$ oracle mask
- (c) In-distribution tracks generalize best; flips dominate failures

The best checkpoint beats the last

- 100k: passive $\rightarrow$ timeout
- 300–500k: moves but flips
- **900k: 0.30 goal (peak)**
- 10^6 : regresses to 0.20
- $\Rightarrow$ ship the **best**, not the final

Steps	Reward	Goal	Flip
100k	−187.9	0.00	0.00
500k	−8.5	0.00	1.00
700k	+18.5	0.10	0.90
900k	+47.0	0.30	0.70
10^6	+35.6	0.20	0.80

oval track, seg mode, 10 episodes

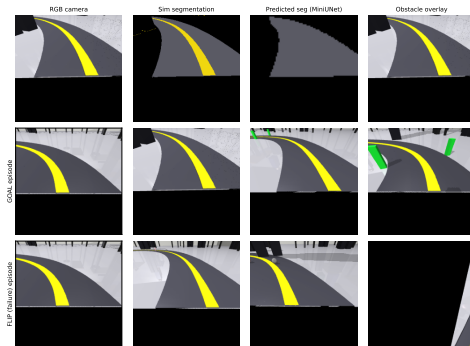
Perception is not the bottleneck

- Same policy, two perception sources
- mask oracle vs seg learned
- **Identical goal rate** 0.20
- Reward drop small: 37.9 $\rightarrow$ 30.1
- $\Rightarrow$ sim perception gap **closed**

Perception	Goal	Reward
mask (oracle)	0.20	37.9
seg (learned)	0.20	30.1

U-Net trained to mIoU 0.881;
palette matches the simulator.

Qualitative: perception pipeline and failure mode



- **Top:** RGB, GT mask, U-Net pred., overlay nearly identical
- **Middle:** a successful goal episode
- **Bottom:** roll-over, the dominant terminal event

PART 3

Discussion & Conclusion

The remaining gap is physical, not perceptual

- Flip rate 0.70–1.00 across all tracks
- Light shell, ω up to 3 rad/s
- Sharp turns / contacts tip past 60°
- High variance $\Rightarrow$ near stability edge

Design fixes

- Lower $\omega_{\max}$, cap angular accel.
- Lower center of mass
- Widen the wheelbase

Limitations and next steps

- Roll-over stability must be solved before hardware
- 10^6 steps is modest for pixel-based control
- Evaluation still uses simulator-rendered RGB
- Next: dynamics regularization + best-model selection
- Then: real-camera **domain randomization** of the segmenter

Conclusion

- Discrete Q-Learning $\rightarrow$ vision PPO in MuJoCo
- Perception gap **closed**: learned mask $\approx$ oracle
- Bottleneck identified: physical **roll-over**
- Ablation turns one run into **design guidance**
- Twist interface is **ready** for a real robot

Thank you

Questions & discussion

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github.com/RodriFMG/RobotinRL